Pseudocode

Main Program

```
BEGIN
CREATE empty dictionary students
REPEAT
    DISPLAY main menu
    INPUT choice
    IF choice = 1
        CALL Add Student
    ELSE IF choice = 2
        CALL Search Student
    ELSE IF choice = 3
        CALL Update Student
    ELSE IF choice = 4
        CALL Delete Student
    ELSE IF choice = 5
        CALL Display All Students
    ELSE IF choice = 6
        CALL Calculate Average
    ELSE IF choice = 7
        CALL Find Highest Scorer
    ELSE IF choice = 8
        CALL List Students by Department
    ELSE IF choice = 9
        CALL Count Students
    ELSE IF choice = 10
        CALL Display Departments and Subjects
    ELSE IF choice = 11
        DISPLAY "Thank you"
        EXIT LOOP
    ELSE
        DISPLAY "Invalid choice"
UNTIL choice = 11
END
```

Add Student

```
BEGIN
INPUT roll number
IF roll number exists
    DISPLAY "Student already exists"
    RETURN
END IF
INPUT name, department, registration number, date of birth
CREATE empty subjects, marks and attendance lists
INPUT number of subjects
FOR each subject
    INPUT subject name and marks
    ADD values to lists
END FOR
FOR each subject
    INPUT attendance
    ADD attendance to list
END FOR
INPUT email, phone and address
CREATE student dictionary
STORE student dictionary using roll number
DISPLAY success message
END
```

Calculate Average

```
BEGIN
INPUT roll number
IF roll number does not exist
    DISPLAY "Student not found"
    RETURN
END IF
GET marks list
total = SUM of marks
average = total / number of marks
DISPLAY average
END
```